

conditional statement

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hypothesis

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conclusion

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negation

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converse

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inverse

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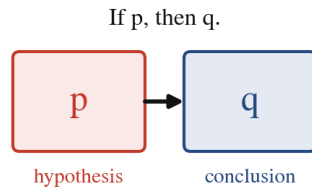
contrapositive

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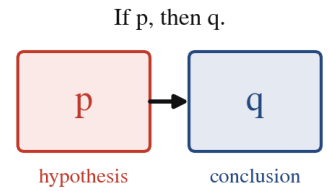
equivalent statements

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The “if” part of a conditional statement — the condition. (p)



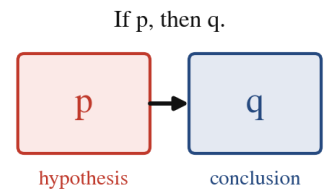
A logical statement with two parts, written in if-then form: If p, then q ($p \rightarrow q$).



The opposite of a statement, written $\sim p$ (“not p”).

Statement: The sky is blue. Negation: The sky is not blue.

The “then” part of a conditional statement — what follows. (q)



Formed by NEGATING both the hypothesis and conclusion: $\sim p \rightarrow \sim q$.

Formed by EXCHANGING the hypothesis and conclusion: $q \rightarrow p$.

Not always true when the original is true.

Two statements that are both true or both false. A conditional and its contrapositive are equivalent; so are the converse and the inverse.

Formed by negating AND exchanging the hypothesis and conclusion: $\sim q \rightarrow \sim p$.
Always has the same truth value as the original.

biconditional statement

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truth value

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perpendicular lines

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inductive reasoning

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conjecture

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counterexample

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deductive reasoning

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Law of Detachment

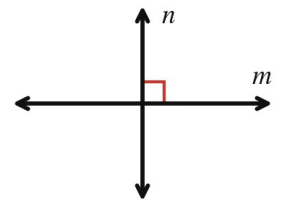
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Whether a statement is true or false.

A statement that contains “if and only if.” It is true only when the conditional AND its converse are both true. Written $p \leftrightarrow q$.

Reasoning that uses specific examples and patterns to form a conjecture.

Two lines that intersect to form a right angle.
Written $m \perp n$.



A specific case for which a conjecture is false. One counterexample is enough to disprove a conjecture.

An unproven statement that is based on observations — an educated guess.

If the hypothesis of a true conditional is true, then the conclusion is also true.

$p \rightarrow q$ is true and p is true, so q is true.

Reasoning that uses facts, definitions, accepted properties, and the laws of logic to form a logical argument.

Law of Syllogism

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Two Point Postulate

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Line-Point Postulate

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Line Intersection Postulate

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Three Point Postulate

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Plane-Line Postulate

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Plane Intersection Postulate

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line perpendicular to a plane

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Through any two points there exists exactly one line.

If $p \rightarrow q$ and $q \rightarrow r$ are true, then $p \rightarrow r$ is true.
(Chain the statements together.)

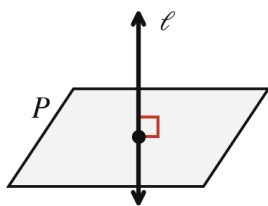
If two lines intersect, then their intersection is exactly one point.

A line contains at least two points.

If two points lie in a plane, then the line containing them lies in the plane.

Through any three noncollinear points there exists exactly one plane.

A line that intersects the plane in a point and is perpendicular to every line in the plane that passes through that point.



If two planes intersect, then their intersection is a line.

Addition Property of Equality

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Subtraction Property of Equality

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Multiplication Property of Equality

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Division Property of Equality

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Substitution Property of Equality

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Distributive Property

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Reflexive Property

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Symmetric Property

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If $a = b$, then $a - c = b - c$.

If $a = b$, then $a + c = b + c$. (Add the same amount to both sides.)

If $a = b$ and $c \neq 0$, then $a \div c = b \div c$.

If $a = b$, then $a \cdot c = b \cdot c$.

$a(b + c) = ab + ac$.
Used in proofs to justify expanding or factoring.

If $a = b$, then a can be substituted for b in any equation or expression.

If $a = b$, then $b = a$.
If $\angle A \cong \angle B$, then $\angle B \cong \angle A$.

Any quantity equals (or is congruent to) itself:
 $a = a$; $AB \cong AB$; $\angle A \cong \angle A$.

Transitive Property

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proof

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two-column proof

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theorem

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paragraph proof

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flowchart proof

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Right Angles Congruence Theorem

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Congruent Supplements Theorem

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A logical argument that uses deductive reasoning to show that a statement is true.

If $a = b$ and $b = c$, then $a = c$.
If $\angle A \cong \angle B$ and $\angle B \cong \angle C$, then $\angle A \cong \angle C$.

A statement that can be proven true. Once proven, it can be used as a reason in other proofs.

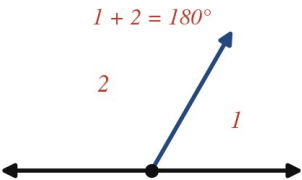
A proof with numbered statements in the left column and the reason for each statement in the right column.

Statements	Reasons
_____	_____
_____	_____
_____	_____

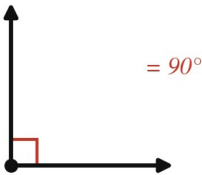
A proof that uses boxes and arrows to show the flow of the logical argument; reasons are written below the boxes.

A proof written as sentences in paragraph form; every statement still needs a reason.

If two angles are supplementary to the same angle (or to congruent angles), then they are congruent.



All right angles are congruent.



Congruent Complements Theorem

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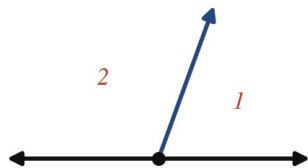
Linear Pair Postulate

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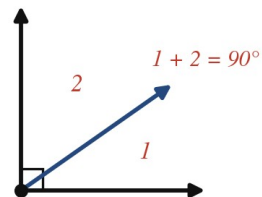
Vertical Angles Congruence Theorem

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If two angles form a linear pair, then they are supplementary.



If two angles are complementary to the same angle (or to congruent angles), then they are congruent.



Vertical angles are congruent.

